

ActInf GuestStream #016.2 ~ Mark Solms

"Consciousness as Precision Optimization"

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hello and welcome everyone this is actin flab guest stream number 16.2 it is june 11 2022. we're here with mark solms this is the second part in our discussion we had an awesome part one today we are going to have a recap on some of the points that were addressed in the first part in 16.1 and then we're going to head into some unexplored territory so mark thanks so much for joining for these sessions and please take it away thanks daniel glad to be here again um i i was just testing out sharing my screen a minute ago and it seems as if i'm incapable of showing you my face and sharing my screen at the same time so take a good look at me you're not going to see me for a while uh here comes my screen please god there we are great uh and here comes my pointer there we are and here's my title the emphasis is on physiological and philosophical considerations rather than computational ones so i'll start with a brief recap i made a few claims um and i didn't get to the end of the this list um so i will just remind you what those claims were and then i will pause when i get to the point that we stopped at last time which was here that affect is an extended form of homeostasis started with the claim which really is the main claim of my of my presentation it is that affect feeling is the foundational form of consciousness and that it is intrinsically conscious um i quoted freud who more than anyone else uh introduced the notion into mental science that mental processes are not that much of our cognitive processing goes on unconsciously even he made this point the point that i'm making which in his words was that it is surely of the essence of an emotion that we should be aware of it that is that it should become known to consciousness thus the possibility of the attribute of unconsciousness would be completely excluded as emotions feelings and affects are concerned and i emphasize the word feelings uh because there's a whole lot of reasons why some people claim that there are such things as unconscious emotions and unconscious affects this so just to be absolutely clear what i'm talking about here i'm talking about feeling and you can't have an unconscious feeling in other words you can't have a feeling that you don't feel so whatever might be meant by unconscious emotions and

unconscious affects um that they can't apply to feelings they can't be unconscious feelings but uh that might sound like a semantic point uh just a matter of words that you can't have a feeling that you don't feel that's an oxymoron so i then went on to set out various reasons empirical ones uh why the claim that feelings are or also has a solid empirical basis i drew attention to this discovery that was made in 1949 already that the brain mechanisms for consciousness itself in other words what wakes us up in the morning and and puts us to sleep at night the sort of switching on and off of the lights of consciousness uh that this is the business of the reticular activating system the reticular activating system is the brain system for arousing consciousness for activating uh cortical processes and rendering them conscious i also mentioned the periaqueductal gray which plays a a and also it also plays a crucial part in all of this together with the reticular activating system but i'll come back to that later for now let's just focus on the reticular activating system to give you a sense of what i mean by how these structures their intrinsic function is the switching on of the lights of consciousness uh i i pointed out that you can make a lesion or you can suffer a lesion as small as two cubic millimeters in size uh in the parabrachial complex of the reticular activating system in other words a lesion the size of a match head and that is enough in human beings reliably to obliterate consciousness entirely so i was saying let's look at these structures if we're wanting to understand something of the intrinsic consciousness generating mechanisms of the brain and um i made the point uh that if the cortex which is where we normally shine our light in terms of looking for the neural correlates of consciousness if the cortex where the seat of consciousness then there's an easily testable prediction which is that if you have a case in which there is no cortex then the patient should not be conscious not so therefore i showed you one such case uh this is one representative example of its type uh that here is a patient uh born with no cortex uh the condition is called hydran encephaly here you see in an mri scan of her brain that where cortex should be there is just cerebrospinal fluid and you see that her brain stem is perfectly intact so on the view that this brainstem area is where the consciousness is generated that it doesn't require cortex this is a critical case on the brainstem view she should be conscious on the cortical view she should not and here she is and she's conscious as you can see she wakes up in the morning she goes to sleep at night uh in this sense the the lights are on but much more interestingly uh she's not merely blankly awake uh she her her her wakefulness has a content and the quality and the quality that i'm talking about is affective quality you see how she responds to her baby brother being placed on her lap she responds with some form of pleasure um and so there's a there's a content to this mental state and a

quality to this mental state that she's displaying why that's important is because when we first learned in 1949 that the reticular activating system is prerequisite for all consciousness that there's no such thing as cortical consciousness without brain stem arousal of the cortex though we had the view that the cortex provides the qualities and the contents of consciousness and the reticular activating system merely provides the the quantitative dimension the sort of level of consciousness so it's as if this was a power supply as like a television set needs to be plugged in at the wall the the reticular activating system is the prerequisite sort of booting up of the system but the television set itself uh the cortex where the contents and qualities of consciousness are processed uh could still be claimed to be the seat of consciousness and the reticular activating system merely as a as a power source this is why it was important to point out that this child is not only conscious in some blank sense of wakefulness without content and quality but rather that she displays affects she's emotionally responsive to her baby brother being placed on her lap just as this child who also has no cortex is emotionally responsive here you can see she's responding with pleasure to a stimulus and this is quite generally the case for these children has beyond marcus summary uh of his observations in many many many such children they express pleasure they smile they laugh they show aversion and fussing by arching their backs and crying their faces are animated by these emotional states they build up play sequences they smile they giggle they laugh they show great excitement etc etc etc um so i've highlighted all of those words to show that these two cases that i just showed you with no cortex they are conscious and they are responsive and in particular they are emotionally responsive um and on this i base the claim that these children uh do have a quality uh that they're cut they do have a consciousness and this consciousness does have a quality and it does have content many people are perplexed as to how this could happen uh since they have no cortex how how can they respond to things like their baby brothers being placed in their laps and so i just inserted this slide i don't think i showed it last time just to point out that our sensory end organs here's the example is the eye same applies to to the skin and the and you know to to hearing and taste as well the the the the uh optic nerve projects to the lateral geniculate and from there to the visual cortex but not only to the visual cortex it also projects subcortically to the superior colliculi of the midbrain of the brain stem which is immediately adjacent to the periaqueductal gray which i said earlier i was going to mention again so these children receive information in the brain stem which is not conscious it's not cognitively conscious perception conscious perception is generated in the cortex but unconscious sensory information goes into the brain stem where it is responded to consciously by the emotional

structures that that are the main focus of what i'm talking to you about now um of course many of you and like many of my colleagues will reasonably say well how do you know that there's something it's like to be those children these might just be reflexes or instincts uh these these might be uh the equivalent these kids uh of philosophical zombies in other words they look as if they're conscious they behave as if they're conscious but we can't know that they're actually conscious and so last time i tried to address this objection by drawing attention to what happens in cases who lose great swathes of their cortex and are able because they haven't lost all of their cortex they are able to describe to us what it's like to be them this is one way of getting around the objection that because these kids have no cortex and therefore can't declare their conscious states we can't be sure that they have conscious states so i focus first of all on a case who has a massive lesion of the prefrontal cortex the reason i did that is because this is a favorite part of the cortex for those who claim the cortex is the seat of consciousness like the global workspace theorists they say it all comes together in the prefrontal lobes all of this information that's processed in the posterior cortexes is re-represented or accessed in the global workspace and this this is where the sentient being the subject of the mind comes about and i pointed out last time that if that were the case then again we have a falsifiable prediction a patient who has no prefrontal cortex like this patient of mine a patient who has no prefrontal cortex but he has a sliver of language cortex he if his sentient being was was contingent upon uh the integrity of prefrontal cortex uh he should not have sentient being and so i showed you what it's like uh i talked to him about what it's like to be him he claims to be consciously aware of his thoughts i asked him to imagine something for me to imagine two dogs and the chicken to see them in his mind's eye and then i asked him to count the legs i thought this would be a reasonable test of whether he's actually got conscious mental imagery and please note the person i'm talking to refers to himself as i um so he seems to think that he exists as a sentient being and i ask him how many legs they are in total if you have two dogs and one chicken and to my disappointment he said eight and when i questioned his answer he answered he pointed out that in his mind's eye the dogs had at the chicken so i thought that was maybe not a great joke but it certainly suggested that there was somebody at home and i made the point last time that these patients generally patients with massive frontal lesions that this uh tendency to make pure oil jokes is is is considered to be a rather a common part of the frontal lobe syndrome or the frontal lobe personality and this is part of a bigger story that i mentioned last time which is that these patients are generally quite emotionally disinhibited and why that's important is because remember what we are considering here is the question as

to whether the feelings that you saw in those patients those kids with no whether those feelings could be generated in the brain stem and the claim of the of of corticocentric theorists like joe ledoux for example is that the feelings literally come about uh when they are registered or re-represented or labeled or even named some people claim uh that it's only once you um are able to re-represent this uh this uh uh these subcortical uh survival uh circuits uh in uh a declarative consciousness uh in in prefrontal lobes that this is literally what brings the feeling about and i think it's quite interesting that these patients who have no prefrontal don't have a dearth of feeling which is what you would expect if the machinery that brings feelings about is absent then they shouldn't be able to have feelings but in fact what is generally accepted is these patients have an excess of feelings there's disinhibited emotionality in these patients um my i made much the same argument about uh the the other major cortical area that is associated uh with the sentient self uh and that is the insular cortex of course this is associated above all with the work of bud craig but has been very widely accepted uh that the feeling self comes about um when um when uh once the state of one's uh intraceptive uh body uh is is is uh re-represented in insular cortex and uh so again we have a falsifiable prediction uh if you take a case who has no insular cortex like this patient of demarzios then there should be no sentient self uh present and i showed you this interview where di marzia speaks to him about his sense of self and the patient is perfectly adamant that he exists as a self he speaks about himself as i i you know at the end of this interview ends with dimarzia saying you're aware that i'm aware and the patient b says i'm aware that you're aware that i'm aware you know all and and this patient just like my patient w is not deficient in emotionality uh in fact uh all the basic emotions are present including both bodily affects and emotional affects and sensory affects and um not only that he's also a little too emotional and this is a little disinhibited in his emotionality and this is what we quite generally see with patients uh with incident lesions so again it's very hard to sustain the argument that the self actually comes into being in the cortex now of course those two cases those examples of their kinds patients with massive frontal cortical lesions pace of patients with massive insulin lesions of course they are just examples uh but uh the point is the point i'm making now is that the there's lots of cortex left in those cases the argument is well you know maybe this the sentient subject is generated in the remainder of their cortex and it's a bit of a that's a bit of a circular argument because remember you know the the children who have no cortex um we're told well how do we know uh that that they're conscious they can't report their conscious states i then gave these two outstanding examples looking at the areas of cortex that are most bound up with with a sentient

subjectivity according to cortical theories of consciousness and uh these patients seem to have uh intact a sentient subjectivity but now i'm told well you know the remainder of the cortex might be what's what's generating the consciousness so we can't use only lesion methods there's no way out of that armpass and so i then showed you last time evidence of a different kind i showed you what happens if you stimulate reticular activating system nuclei there again there's a there's a an easily tested prediction the prediction is that if these brain stem nuclei the reticular activating system and periaqueductal gray as i'm claiming if these are the structures that generate consciousness and feeling then stimulation of these structures should stimulate conscious feelings and i showed you that here's a case who had a deep brain stimulator placed into the substantia this this part of her reticular activating system and within five seconds uh it's it's it it generated uh it produced a profound depression the patient was actually suicidal she didn't want to live anymore this is a patient with no psychiatric history the the electrode was placed in her brain stem for the treatment of parkinson's disease and and stimulated the wrong nucleus and that's how this came about 90 seconds after the the stimulator was switched off the depressed feeling disappeared and the patient generously agreed to allow for further stimulation uh in the reticular activating system and out of it and it was only when that particular nucleus was stimulated that that she fell into the depression so this is the kind of evidence and remember again i'm just giving you examples you can stimulate intense affective states by stimulating reticular activating nuclei and periaqueductal gray you get the greatest intensity and the greatest variety of efforts from stimulating there and you get nothing of the kind from stimulating cortex this is a different line of evidence suggesting that feelings are actually generated in the upper brain stem and then i showed you another line of evidence that was positron emission tomography of people in intense affective states here you see research participants in in states of sadness here of anger here of happiness here of fear and in all instances the activation uh is in the brainstem and the circuits arising from it the subcortical circuits arising from it that's what we see the cortex is by contrast largely deactivated so this is a further line of evidence you know an entirely different line of evidence remember we've got lesion evidence then we've got deep brain stimulation evidence here we've got positron emission imaging uh uh evidence that the part of the brain that's generating the feelings is the part of the brain that switches on the lights and this is why i'm claiming that the basic uh the foundational form of consciousness and this prerequisite form is affect affect is the elemental form of consciousness a further line of evidence is pharmacological manipulations if you tinker with the neuromodulators that are sourced in these reticular activating nuclei like for

example noradrenaline or serotonin or dopamine all of which are sourced in the reticular activating system now serotonin is of course regularly um pharmacologically manipulated for the treatment of depression uh dopamine for the treatment of psychosis uh noradrenaline for the treatment of anxiety noradrenaline is sourced in locus ceruleus complex serotonin in the raphe nuclei dopamine at least the one that's important for psychosis in the ventral tegmental area all of these are parts of the reticular activating system if all that this system did was switch on blank wakefulness it might be of interest to anaesthetists uh but in fact it's it is the main target of the drugs of psychiatrists uh who are treating emotional disorders by manipulating the chemistries that are the source nuclei for which uh are in the reticular activating system on the basis of all of that i argued remember that affect is the foundational form of consciousness and it's intrinsically conscious i said that feelings uh we don't turn not only on semantic grounds i'm saying that feeling is the basic form of consciousness the reason i'm saying that is because the basic consciousness generating tissues of the brain the reticular activating system which is prerequisite for the activation of consciousness in cortex and that this uh that these structures generate feeling uh that feeling therefore is prerequisite is foundational for all forms of consciousness i then made if this is a sort of a sidebar um the reason i went into this third point um is that namely that the claim that affect is not synonymous with interoceptive inference in other words that it's not just an interoceptive form of perception as opposed to the extra-receptive forms which have been the major focus of consciousness studies over the last few decades um vision being the main focus i'm saying that affect is not and why i say this is because this is increasingly being argued um in fact it was argued by bud craig himself that affect is just interoceptive the perception of the state of the own body is the equivalent of the perception of the outside world in the more typical form or extra-receptive form of perception that's been equated with consciousness i made the point that there's that there's good reason to believe that affect isn't just a sixth modality of perception um and uh the evidence that i presented for that was of various kinds uh for example that there are affects which are clearly not receptive for example getting a fright uh was being startled um or feeling pain when when when when you you a pin is stuck into your finger these are all extra-receptive forms of stimulation and yet they arouse affective responses so affect clearly is not uniquely interoceptive and then i also made the point that there are many interoceptive perceptual states which are not affective you can feel your tummy grumbling or you can feel your heart beating or you can feel your your your lungs expanding these are not intrinsically affective phenomena so interoceptive perception can happen without affect an affect can happen

without interreceptive so i was trying to draw a line under the idea that affect is just another modality of perception i i i think that by detaching affect from perception uh casts some new light on the hard problem and so that's where i went next uh and i reminded you that david chalmers who coined the hard problem uh said that when we look at um functions uh perceptual functions like vision which was as i said the model example derived from the or grounded upon the assumption that consciousness is a cortical phenomenon cortical vision became the model example of consciousness this was following crick's initiative in the mid-1990s crick's idea was that if we can if we can identify the neural correlate of consciousness in the case of cortical vision then we can generalize from that uh by by discerning the mechanism whereby visual information processing gets turned into conscious vision in the cortex by isolating that mechanism we will be able to uh understand uh the the nature and function of consciousness and chalmers said that that's not true you if you if you isolate uh identify the mechanism of visual information processing like this map here does it doesn't tell you anything about why there's something it is like to see and he he used the well-known knowledge argument of frank jackson the story about about mary the visual neuroscientist who knows everything about all of this and i slightly simplify the story by saying well let's let's imagine that mary is blind even though she knows everything about the functions of cortical vision she knows everything about the mechanism whereby visual information is processed in the cortex because she's blind she knows nothing about what it is like to see and if she were to be gifted suddenly with a normal site then she would learn something completely new about vision she would learn what it is like to see red and blue what blueness and redness etc are like none of which is accounted for by this information um processing flow diagram and none of what she knew about the functional mechanisms of cortical vision that would have prepared her for what it is like to see in other words there's other there's there's something some something else about visual information processing other than the sort of things that mechanistic um functionalist dissections like this uh provide us with uh it doesn't this this mechanistic account doesn't predict what that there should be anything that it is like to see and this is the grounds upon which people like chalmers say that a mechanistic effect an account of the functional mechanism of uh cognitive and perceptual processes it doesn't tell us anything about um why there's something it is like to to to to perform these processes and this is the essence of the hard problem um i said that i thought that this might be because they were looking in the wrong place that visual perception uh and and and all forms of cortical perception um and uh indeed not only uh perception with

learning and cognition more generally that this can readily go on unconsciously that these are not intrinsically conscious processes that it is perfectly possible to see um and to recognize faces and to read with comprehension and even to discriminate colors these are all uniquely cortical processes and they can all go on in the dark as it were in other words you do not have to be conscious of what you're perceiving in order to perceive it and so this this costs a rather different light on thomas's point that all of this information processing can go on in the dark um so you know consciousness uh can't isn't accounted for by our normal uh functionalized mechanist mechanistic way of doing uh cognitive neuroscience so i was saying well that's because they're looking at functions like and functions like learning uh which dude which are not uh these are not intrinsically conscious uh processes um i i i drew your attention to uh this other statement of chalmers is from his famous paper in which he says in summary there is no cognitive function such that we can say in advance that explanation of the function will automatically explain experience and my point was well that's because they're talking about cognitive functions could chalmers make the same statement if we were talking about affective functions i'm saying no he could not i'm saying that we can say in advance that explanation of the function of will automatically explain experience because the function of feeling is to feel it's intrinsically conscious is this is this function unlike vision and perception in general and cognition as a whole none of that is intrinsically conscious but feeling affective feeling is intrinsically conscious you could not understand the mechanism of affective feeling if it didn't account for why it feels like something because that's the whole bloody point of feeling so this question of chalmers is why is the performance of these functions accompanied by experience why doesn't all this information processing go on in the dark free of any inner feel i'm saying that that question is perfectly reasonable when asked of these cognitive functions which are not intrinsically conscious that kind of information processing can go on in the dark free of any inner feel but that is not true of feeling of affective feeling and so uh my claim and this is where we got to last time this is uh now i'm now starting with new arguments my claim is that if we can identify the functional mechanism of affect of affective feeling then we will be able to explain why this sort of information processing doesn't go on in the dark so i hope that that's clear that's kind of like my main point my main point is we've been we've been looking in the wrong place and we've been looking to cortical vision and cortical perceptual and cognitive processes in general in order to isolate the neural correlative consciousness and chalmers has said well it doesn't work you can isolate the mechanism whereby this sort of in these sorts of information processing goes on and it doesn't tell you

anything about why there is something it is like to see etc and so my argument is yes that's true of those processes but it's not true of affect affect uh the the the remember what i've said to you in my summary now today that this is the most concentrated consciousness generating tissue that there is the reticular activating system and periaqueductal gray is where the lights are switched on um and that the you need uh you you can uh switch those lights off with as small a lesion as two cubic millimeters in extent so this is the place where we should obviously have been looking in the first place uh in order to understand to identify the neural correlate of consciousness and to enter and to understand its mechanism much more important than that um is the fact that the kind of consciousness it generates uh in that namely feeling is it is it is an intrinsically conscious mechanism that feeling uh it wouldn't exist if it wasn't felt and so unlike vision and and and learning and cognition in general which does not have to be felt uh affect does have to be felt that's the whole point of affect so i'm saying that if we could understand the mechanism of affect in other words the mechanism whereby feeling comes about then we might make some progress with this hard problem of consciousness so remember that's how far we got lost time and now i go on to and i'm now going to slow down a little bit because i'm now going to argue try to identify what the functional mechanism of affect is but i thought perhaps before i do that i should pause for a moment in case there any questions arising from my summary of my argument so far carry on thank you mark okay thanks so um you see i'm saying affect is an extended form of homeostasis and by the way this is not my um argument this is an argument that was first uh to my knowledge first introduced by jacques pancep uh in the 1990s uh subsequently uh uh popularized by antonio de marzio so i'm just summarizing my version of what is in fact now a good 25 year old argument homeostasis is probably the most basic biological mechanism it's it could be said that homeostasis is what enables living organisms to resist the second law of thermodynamics in other words uh it is what enables them to be living organisms um they don't just dissipate uh they remain in an organized form basis of this is homeostasis rather than just explore all possible states we living things have to remain within highly specific states these are called the settling point or the set point the viable ranges of the organism and this applies across multiple different parameters let me use the example of core body temperature for those of us who think in degrees celsius you know we have to remain between 36 and a half and 37 and a half degrees celsius that's where we need to be if we if we deviate too far from that very narrow range then we are at risk of dying and that doesn't apply only to temperature it applies to water uh to oxygen uh to salt to sugar to blood pressure to all sorts of things about our our

bodies they have to remain within very narrow ranges that are viable with the preservation of our of our living state if we move outside of those ranges in other words if we explore all possible uh states uh then we die so we have to work against that dissipative trend as i said that is this entropic trend uh we have to resist entropy uh and and and remain within our viable states a very narrow specific ranges of viability across these multiple um dimensions of of our physiology this is why i say that homeostasis is what keeps us alive now a deviation from that settling point then becomes a demand for work the organism has to do something to return itself to its viable bounds and that's the basic mechanism of homeostasis that's how all of those autonomic functions i was talking about earlier um like core body temperature and blood blood gas balance and so on that's how they work uh if you're moving outside of your range you have to do something to return yourself into the range that is what homeostasis is it's the it's the mechanism that returns us back to our viable range what i'm saying is that affect is an extended form of homeostasis and how it extends homeostasis is that when we move out of our viable range we feel it we feel an unpleasant quality that means i am moving out of my viable range and by contrast if you're moving back towards your viable range you have pleasurable feeling this seems to be the basic function of affect it enables the organism to know how it's doing in terms of its organismic viability the organism feels when it's moving outside of its viable range as displeasure which means this is bad and it feels moving back towards its viable range back towards its ideal settling point as pleasure which means this is good this predicts my survival this predicts my demise so feelings are rooted in a value system in other words that there is something good and there's something bad and what is good is to survive and as it happens to reproduce and what is bad is to not do so this is of course the basic value system that underwrites all of life this is the value system that underwrites natural selection so the so feelings uh are rooted in that value system and what they do is they enable the organism subjectively the organism itself to know whether it's moving out of its viable bounds in other words whether it's doing something bad within that value system or something good within that value system that's what feelings do they enable the organism to know how they are doing within that value system now why why does this get added to homeostasis because not all homeostasis is felt most homeostasis is entirely autonomic i was mentioning earlier uh blood pressure for example when you move out of your viable range your heart rate changes and your the the the your blood vessels dilate um in order to um return you to your viable blood pressure you don't need to know in fact you do not know anything about it uh in fact it's clinically notorious how blood pressure regulation works uh because you can be way

out of your viable bounds and know nothing about it and i'm just using blood pressure as an example there are many many many ways in which your autonomic uh nervous system maintains you within your viable bounds without you knowing anything about it so why do we need to feel it uh well the the what feeling adds is when you're in a situation of uncertainty and a situation where you do not have a readily uh pre-prepared reflex which returns you to your viable bounds so you don't need to feel how you're doing if you don't have to make any choices if you have automatic predictions which return you by reflex uh like in the case of blood pressure regulation uh to your viable bonds then you the sentient being um have no part to play in the process uh what what where feeling comes into its own is where the organism finds itself in a state of uncertainty for example in a novel situation for which it's it's uh innate um pre-wired as it were reflexes uh have no have no uh preparedness um you find yourself in a state of surprise and uh now what you're going to do uh the the the if there's no reflexive solution available then feeling enables the organism to make choices choices have to be rooted in a value system there has to be a good choice and a bad choice otherwise it's random whether you're doing the right thing or the wrong thing announced to you by how it feels so this enables voluntary action i really must emphasize that this enables you to choose it enables you to decide for yourself what to do things are getting worse so i'm going to change my mind things are getting better i will stick with policy this is working so these are not these are not hardwired these are not innate predictions these are choices made on the fly by the organism here and now and they enable the organism to survive in states of uncertainty in other words in unpredicted situations in other words in novel situations and god knows there are many of those in life just to put flesh to those bones let me give you an illustrative example normally respiratory control is autonomic resp you don't need to make any decisions uh about breathing it just happens automatically so you automatically remain within your viable bounds in terms of the ratio of oxygen to carbon dioxide but that's only when the normal autonomic reflexes manage the situation because you are in predictable circumstances now imagine that you are in a carbon dioxide filled room now suddenly you move out of your viable bounds without bounds and breathing normally normal reflexive uh regulation uh of the of of your respiratory system doesn't work uh because ordinary breathing uh in this in this carbon dioxide-filled room that is going to rapidly lead to your demise so now you've got to do something you've got to get you've got to you've never been in a burning building before let alone this particular one so you have no you have no uh reflex or instinct no innate prediction as to what to do so you feel your way through the situation and please note at this point at the

point when you find yourself in this unexpected situation this is where this is where your need for oxygen becomes conscious so this is a very important point an otherwise autonomic function uh now becomes a conscious function uh you feel what we call air hunger or suffocation alarm um and you now you move uh about in this building uh deciding which way to go remember you have no you know prior knowledge of what to do and it's only on the basis of how it feels so if for example you go upstairs and the oxygen supply diminishes you feel worse air hunger if you go downstairs and there there's a greater provision of oxygen then you feel better you feel relief from the suffocation alarm and so the feelings tell you whether what you're doing is working or not and so your choices are based on feeling and so you are able to feel your way through this problem and survive this is not a small advantage the ability to survive in unpredicted environments is an enormous adaptive advantage and so that we believe is what the function of feeling is that's why in this narrow example uh why respiratory control suddenly becomes conscious uh that it that it uh uh dramatically intrudes on consciousness suddenly your need for oxygen uh and the purpose of this is to enable you to calibrate your choices uh to to to to change your mind uh about your policy your current policy on the basis of whether it's working or not uh the which is which is exactly what feeling announces for you in the absence of feeling you would behave randomly and one in a million will do the right thing and that one will survive and reproduce and the rest of you have had it so this enables us to make choices within our own lifetimes we don't have to let natural selection do it within our own lifetimes we can adapt to unpredicted situations of course once you've done that you then also can learn from the experience within your own lifetime and so the next time you find yourself in a burning building you might on the basis of of of of learning from experience you might have somewhat less uncertainty about what to do again let me just make sure that i'm getting across my main point because this is the mechanism of feeling this is the function of feeling this is what feeling does this is why the organism must feel it now remember what um chalmers had said uh that you know why doesn't all this information processing going in the dark and so on that was built upon an earlier argument by tom nagle who said an organism has conscious mental states if and only if there is something that it's like to be that organism something it's like for the organism then he went on to say if we acknowledge that a physical theory of mind must account for the subjective character of experience we must admit that no presently available conception gives us a clue about how this could be done so this is what i'm trying to do in this talk i'm trying to give us a clue about how this could be done as this is nagel's way of formulating the hard why and how is there

something it's like to be an organism something it's like for the organism my point is that that question only makes sense in relation to things like visual information processing which it does not have to be something it is like to see there does not have to be something it is like uh to discriminate red for blue you can do that unconsciously the cortex can do that automatically go on in the dark but would you would charmers even have asked i mean nagel even have asked this question if we were talking about feeling i've just explained to you what the function of feeling is and it makes it kind of absurd to ask why and how is there something it is like to be an organism something it's i hope that you can see what i mean why and how there is something it is like to be an organism something it is like for the organism has everything to do with feeling feeling is is what it is like to be an organism how you're doing as an organism how much longer are you going to remain in existence as an organism that's what feeling is like and and and and this is why and how uh it exists so if we can get to what the physical mechanism going back to what to what uh to what nagel was saying here he said if we acknowledge that a physical theory of mind must account for this something at his likeness then he's saying we have to admit that we currently have no clue that i'm saying it's because we've been looking in the wrong place i think if we look to feeling and we seek a physical theory of feeling this will account for the subjective character of experience and we will make some progress uh this will provide us some clue about how we might uh go about solving uh this heart problem now um my next point oh so let me pause at that point uh having said that affect is an extended form of homeostasis that's my next claim uh let's see if anybody wants to argue the toss or make a comment or ask a question yeah so stephen there's no need to raise hand after each point we'll take any questions so go for it steven and then dave if you have anything thank you yeah thank you no really really enjoying this um i'm i'm interested in this awareness that leads to the homeostatic um sort of correction and if it's where the awareness is with the actual cells or organs struggling in some way to act or enact upon the surprisal so be that in the environment that they're encountering at the different scales so this this issue of actually acting or enacting and when that action and inaction can functionally correlate up the nested hierarchies then it's those actions or the the inability to act in the way that cells would like that then is becoming the signal of what or where action is fading rather than necessarily a signal of that actually contains the information inherently in it i wonder what your thoughts would be on that yes i agree if i'm understanding you correctly i agree with the emphasis on action um so i'll slightly run ahead of myself if i say this obviously a homeostatic deviation can also be construed as a prediction error in other words what your what you're doing has not

led to the outcome that was expected and there are two ways of correcting prediction errors you can either you can either change your prediction or you can change what you're doing in order to bring about the prediction that you had originally in other words when it comes to when it comes to prediction error you can either update the prior and have a posterior prediction or you've got to do something differently in order to confirm the prior prediction so that's where the emphasis is on action now why this is so important in relation to homeostasis is that you can't change your prediction about what your viable bounds are you know if you expect to be between 36 and a half and 37 and a half degrees celsius on the basis of acting in a certain way in other words firing an autonomic reflex and you then find yourself to be at 39 degrees celsius you can't say okay my posterior prediction is that by doing this i'll be 39 degrees celsius because if you do that you're on the you're you're rapidly on the road to death so so the emphasis so the prediction error has to be corrected um on the basis of changing your action policy doing something different in order to confirm your prior prediction so so the emphasis is very much on action when it comes to these organismic predictions these these phenotypic um the viable states for your phenotype can't be changed uh you you you can't just up you can't just change your mind about what you expect will flow from your actions you have to change your actions in order to bring about the expected or preferred state of your phenotype thanks can i just add one piece to that i think that's really helpful is and also in terms of like multiple scales going down to smaller scales so for instance if i actually feel heat in my body and my i my body might be my cells are trying to act to find a better state that can be seen as oppressive and claustrophobic if i then read that as being me getting into an uncomfortable state it could also be me basking in the sun on a beach you know in the same way that if i taste something that's very sour there's a there's some sort of action at the cell or the or the organ level reacting to that sourness or that sweetness which could be like a nice uh confectionary sweet for a child you know like one of these gob stoppers or it could be quite a problem so i i think i like what you're saying with action i'm almost wondering whether that that action piece the inaction is the prediction error in terms of how it can go up in terms of these these nested hierarchies of of physical scale yes um so again i must be careful not to run ahead of myself too much i mean inevitably you know you must ask whatever question comes into your mind at any point but i i i'm aware that as i go down this list of of of arguments i'm going to be addressing those points so i don't want to make all of those arguments in one go so i will just say that for now that that what you've just asked has first of all a lot to do with the fact that we have multiple needs that there's not only one need that they have to be balanced in

relation to each other and also very importantly that we're talking about a predictive hierarchy this is what you're speaking about with reference to scale uh that what applies uh at the sensory periphery uh and what applies uh at the core of the predictive hierarchy um have different implications for the way that that affect works so i'm just saying those very kind of vague and abstract things for now and i hope that the picture will become clearer as i proceed thank you mark uh yes dave go for it yeah i don't know how to state this clearly um this is some thinking that i i got into from listening to a very recent interview you did with um a young man in another part of cape town i'm talking about perception and i there i think there and maybe this has already been done i just haven't run across the research that there's a larger framework in which notions of what can be perceived what can be perceived consciously what is a sensory modality what we don't call a sense it seems that there are on the one hand conditions we've been calling those homeostatic conditions that you can't do anything about uh your body either adjusts the blood pressure internally or it goes haywire but you can't be aware of it so we don't think about this as something that we can consciously perceive and we don't call the mechanisms that are sensitive to blood pressure and uh core temperature and other things as matters of sensation that is a perception matters of consciousness on the other side the other side of the mapping there are single object uh influences a a mechanism that is sensitive only to blood pressure presents another reason not to call this a sense um an eye spot just tells a certain kinds of worms there's light around here or there isn't light around here no directionality just it's daytime or it isn't i'm under the mud or i'm not under the mud [Music] whereas even a little more information more kinds of information qualify this more as a sensory modality and in things like hearing and vision and smell we have this very rich many to many there's many things you can do about about many things you can do about a burning smell and there are many kinds of odors out tie into anything you're you've been thinking about yes um so the first thing that you're touching on there the fact that uh homeostasis is only part of the story uh in fact the the the the example i gave earlier uh of the of the person who finds themselves in a carbon dioxide filled room in a burning they need to maintain uh their homeostatic blood gas balance um but there's nothing that their autonomic nervous system can do about that they therefore need to turn uh the to action in the outside world so when i said that the person starts moving about upstairs and down and then feels um uh is this working or is this not is this good or is this bad that we call allostasis uh it's it's it's acting in the outside world in order to return myself to my homeostatic palms so that's the first thing i just wanted to make clear i know it's implicit uh and i know i'm telling you something that you

know very well but i just want to make that explicit for for for uh our participants that um what the importance of the outside world um for uh these internal uh bodily states um the the the the bridge from homeostasis to allostasis is how we conceptualize that so then it moves to the next point which is that think about those kids that i showed you earlier who have no cortex and all they're capable of feeling is their affect and they don't they can't they can't consciously think i feel like this about that in other words they don't know what the about that is they they just feel things uh and they can't include within their consciousness of the feeling uh what the thing in the outside world is that contextualizes that feeling the context doesn't become conscious only the feeling itself to be able to extend the feeling onto the context to be able to say in effect i feel like this about that incorporates the context within terrain of uncertainty where one's navigating the ones palpating the uncertainty in order to make choices not only on the basis of blind feeling but also on the basis of what kind of object brings about this change in my feeling and what kind of object brings about that for all of that to be incorporated within the realm of consciousness i think is a further leap and clearly another enormous adaptive advantage so the first leap is just to be able to feel the consequences of your actions the second leap is to be able to picture as it were uh your those actions within the sphere of consciousness uh and uh and in this way to bring the context into consciousness and extend the realm of choices enormously thank you mark and i'll just chime in with a third question so you mentioned homeostasis is only part of the story and in the chat brock asked what part of the story is hormesis i don't know that word so a small stress inducing like some kind of benefit over the integrated time horizon like an exercise stress that leads to improved strength you mentioned allostasis kind of anticipatory movement towards a set point but where does um for example induced stress mild stress and recovery play into this extended homeostatic or generalized homeostatic framework so um sorry i i didn't understand what you meant so you know i think again when one gives um an overview of a of it of an argument like this one always oversimplifies what one sort of has to simplify and you know when i say that homeostasis that you know you always have to confirm your prior prediction that's not entirely true there are also ways in which the the homeostatic range can be extended uh the homeostatic ranges i mean like for example um you know what i was saying about oxygen um you you could you can if you're a diver you know you can learn how to how to hold your breath and how to manage the stress uh of being out of your viable blood gas range uh in in a way that a naive uh a person like like like i uh cannot so they are they are mechanisms uh whereby uh these these things can in very very narrow limits uh these things can be changed

but i think the emphasis there has to fall on the narrow limits ultimately there is there is an utter limit to your viable range you know this is the driving mechanism of the story that i'm talking about here thank you of course so much more to say and learn and add but please carry on well you can see highlighted on the screen now my next point and it might seem like a small point it might seem like an obvious point that complex organisms require multiple homeostats and in fact i've more or less said this already but i want to make explicit um and here is the same slide that you saw a moment ago where i've just added at the point that i've now that i'm now making the the the the importance of this point is that we have multiple needs which must be categorically distinguished from one another in other words we can't have a continuous variable called need that that variable has to be factorized across a number of different categories so let me be clear what i'm talking imagine you have if we quantify on a continuous how much deviation there is from where you need to be let's say okay i've got um i've got six out of ten of first and i've got four out of ten of sleepiness then all i that means i've got a 10 out of 20 of total need and so all i need to do is sleep i don't need to drink if i if i can generalize from that little example the the point that i'm trying to make is that no that's not true you have to sleep and you have to drink and you have to eat and you have to defecate and you have to breathe you can't just submit all of these and and bring down the total number if you did that you would die uh we we have to recognize that each one of these needs has to be met in their own right um and that means we need to know we can't just have a total variable called need we need to know which need are we talking about and this is why they have to be treated as categorical variables in other words this here we're talking about how much how much water i'm lacking here we're talking about how much oxygen i'm lacking here we're talking about how much sleep i'm lacking um so that they can be so that the category of need uh that the organism finds itself in uh can be addressed uh appropriately the appropriate category needs to be addressed uh and now i know what i'm saying sounds absolutely obvious but why it's important is that it's because of this because of the point i've just been making that hunger feels different from thirst and thirst feels different from suffocation alarm and suffocation alarm feels different from sleepiness and sleepiness feels different from pain and so on because each one of these feelings with the quality of the feeling tells us which need is at issue categorical variables are distinguished qualitatively so we're not only talking about valence goodness or badness and arousal how much or how little we're talking also about qualitative categories um and i think that's important when we come back to the whole point of what i'm talking about namely qualia you know namely um the the qualitative stuff of um

of of what it is like uh to do anything be anything and that those those qualitative distinctions it's of the essence of consciousness uh that that it is qualitative it has qualitative differentiations it's not just some uh a total quantitative some total continuous matter of need it's a matter of different categories of need which therefore are different qualities of need and i think that's important in terms of understanding what an affect is an affect is a state of the organism registered by the organism in other words it is intrinsically subjective it has intrinsic value valence goodness and badness which has existential consequences for the organism and in addition to that it has quality it is inherent in the nature of affects uh that they are qualitatively distinctive from each other and i hope and i hope that this very simple functional mechanistic reasoning that i've given you as to why that needs to be the case why that must be the case um helps to to make clear why affects take the form that they do remember what we're doing here is addressing nagel's question about you know why is there something it is like to be an organism and how does it come about that there is something it is like to be an organism for an organism that something it is likeness is not just a valence goodness and badness but also a qualitatively differentiated state subjective state of so i think that by by reducing the mechanism of affect to these essential features we begin to see it takes the form that it does why it is and necessarily conscious something in the sense that i'm the rudimentary sense that i'm defining consciousness so that's why this point is important uh that's why i wanted to make the point clearly that complex organisms require multiple categorical homeostats they need to be treated as categorical variables they need to therefore be they need to be qualified uh each of these different factors of need each of these different categories of have to be qualitatively differentiated from each other and this is a i think the ground zero of where qualia come from now i'll go on to my next point which is that these different categories of need must be prioritized again i think this is a this is in a way an obvious point but it's but i want to just dwell on it for a moment uh to to to to draw out some of the some of the implications of this i said earlier that our autonomic homeostats run well autonomically most of the time in other words there are automatic automatized which we call reflexes that are when the organism moves out of its viable bounds then it fires these reflexes which bring it back into its viable bounds and then i told you that not all of our needs can be dealt with that way some of them as we move outside of the range of what the autonomic reflex can achieve like for in when it comes to thermoregulation you start to perspire if you get too hot this is a reflex you start to breathe more shallowly and rapidly panting in other words that's a reflex these are autonomic responses to overheating then they reach a certain limit um you know that

you that they they that they haven't cooled you down adequately so then you've got to do something you want to do something like leave the kitchen and that is when that is that is uh what i've just said applies to one of your multiple homeostats in other words one of them now requires allostatic action you now need to do something in the outside world and clearly you can't do something in the outside world in relation to all your needs simultaneously so all the time you're sliding you know your your your your hydration your water in relation to salt uh content of your body sliding all the time it doesn't mean that you're thirsty all the time uh you're you're burning up the the sugars in your adipose tissues all the time that doesn't mean you're hungry all the time so the question is you know what gets prioritized and there must be some prioritization because you can't do everything at once when it comes to voluntary activity in the outside world you have to prioritize there's an action bottleneck and the point i'm making here is first of all just simply that that you do have to prioritize and secondly that what is prioritized is what becomes as i gave in the example of suffocation alarm that that business carries on unconsciously until it becomes prioritized at that point it forcibly intrudes on consciousness my priority now is i need oxygen and that's what you feel the other needs this is the the other point i'm making the other needs don't disappear at that point they continue to be regulated but they are regulated so the prioritization of a need brings that need into the realm of palpating uncertainty in other words into the realm of feeling your way through the problem and uh the this i think is an important uh mechan an important part of how of how feeling works so let me here remember i said i was going to talk about the periaqueductal gray so and i was going to tell you how plays in as important if not more important role in the basic machinery of consciousness of the upper brain stem than the reticular activating system the periaqueductal gray which is just a 14 millimeter long structure columnar structure around the central canal of the midbrain all of our homeostatic mechanisms in other words all of the multiple homeostats and we have many all of the multiple homeostats all of them send their residual error signal to the periaqueductal gray the periaqueductal gray is like a final common pathway of all of these homeostatic error signals and it seems that this is where the prioritization must be must be going on the the the the the the the determining which of the current error signals is the most salient then gives rise to and this is an argument i must again make clear this is not an argument of this is not my own novel insight this was argued on comparative anatomical grounds by beyond marker uh in in his in his in in a brilliant paper uh in behavioral and brain sciences uh so he he recognized that the periaqueductal grey together with the superior colliculi and the midbrain locomotor region they form what he called a

midbrain selection triangle in other words this is where the affect the need that is to be prioritized is selected and this gives rise to a feeling uh the feeling being as in the example that i gave with suffocation alarm the need that is prioritized the need that is currently most salient the need that is going to color your consciousness qualify your consciousness in terms of its most rudimentary property in other words the feeling the affect what state what organismic state am i in i'm in a state of respiratory distress i'm in a state of suffocation alone that's what i feel why it's because there this is where choices need to be made this is where the creature needs to feel the the person in that example needs to feel their way through the problem the other needs remain but the other needs the non-prioritized needs are not raised to the level of feeling and and i hope it's why and we need to understand mechanistically how does all of this work and that leads to my next point so before i go to that next point let me just pause again in case there are any comments or questions about this prioritization function performed by the periaqueductal gray which is what determines what affective state you're going to be in from one moment to the next in other words which need uh is which need is going to is going to qualify your your affective state thank you mark uh dave then stephen yeah i just want to throw out there um in sylvan tomkins later work that is after 1970 he came to elevate reset to the status of a fundamental emotion and i think it's very relevant to this the notion of suddenly shifting from one emotion or one conscious activity or world attitude to another uh i i i believe uh yuck poncep drew pretty heavily on tomkins and the reason that he doesn't cite him in a lot of detail is that he accepted so much of it kind of like what you said about uh i agree almost totally with um oh the the the britisher whose father's from india i'm sorry i'm blind can i understand you had a dialogue with him just a few months ago um am i completely wrong about it real seth are you talking about yes thank you am i completely wrong this just just did this was thompson did thompson simply agree no no i agree with that completely tomkins uh uh the reason that pancetta doesn't cite him uh or barely ever he cites him usually in in in the beginning of a kind of a general discussion you know just just to say this is the tradition i belong to and then and then carries on building on on his shoulders i agree with that very much thank you stephen yeah i liked uh when you talked about palpitate uncertainty um and the salience around that so that that's a nice term and i think that relates to like what is going on so to speak um so what about what's going on and then now what happens next and i'm wondering though as well if the idea of where which i think is often left out there's a lot of categorization of what things are but where is it happening that makes a difference in terms of how significant it might be in terms of prioritization so i was wondering how much the risk of

the awareness some sort of this equilibrium happens um could be featured in in this and maybe that's where the physicalization um and the spatialization becomes much more significant than when you're just dealing with the brain yes so well um all of these homeostatic mechanisms of course the the physiological processes that they regulate are widely distributed but the control centers of these are for the most part uh in the brain stem broadly defined um and i say broadly defined because there's some that are in the medulla oblongata and pons but there's some that are in the midbrain there are some that are diencephalic um for example i mean just mention the most outstanding example the hypothalamus is full of homeostatic control centers but uh you know so the so so hypothalamus um circumventricular organs um the uh parabrachial complex um the area postrema the nucleus solitarius these are all homeostatic control centers but of fundamental importance and i i think it's not sufficiently recognized although mac has certainly made much of that all of these nuclei that i have just enumerated they in turn project to the periaqueductal gray the periaqueductal grey is uh in my view a metahomoeostatic metahomeostat it's sort of the the the the the this the control center of the control centers uh and that the the essential function there is a prioritization function what's extreme and and by being able to physically locate uh these mechanisms bearing in mind what where i started i i started my answer to your question by pointing out that ultimately these homeostats regulate physiological processes which cannot be localized they are distributed processes part excellence but the center the control center can be localized and those control centers are there are numerous uh ones and they in turn they all they all send their residual error signal to periaqueductal grade that that is really important now what that enables us then to do is to test models like the one that i'm describing to you what happens uh when one or another of these individual homeostatic control centers is lesioned what happens when the periaqueductal gray as a whole is lesioned what happens i might as well just insert that here is you get a persistent vegetative state in other words uh you get that condition that um magoon and maruzi led us to believe uh uh is is is a theoretical possibility in other words blank wakefulness remember i said earlier in my summary of what i said last time i spoke that the idea that the reticular provides merely a quantitative level of consciousness and not any quality or content that that fictional state of affairs of blank waveforms the thing that the reticular activating system was supposed to be producing which as i've showed you is not the case because all of those different lines of evidence that are summarized show that what the reticular activating system is producing is anything but without quality and without content that this is in fact the foundational form of the qualities and contents of consciousness namely the different affects if

you lesion the periaqueductal gray then you get that fictional state that artificial state of blank wakefulness in other words uh the these patients um show they show non-responsive wakefulness they still have the autonomic sleep waking cycle in other words they wake up in the morning and go to sleep at night like those those hydrant and cephalic kids that i showed you but unlike those kids they show no emotional no affective no no no response uh to their situation and no intentionality either and so you know that's uh that's what you would expect uh would occur uh if the periaqueductal gray is where the affects are actually being generated because remember that's what i'm saying i'm saying that the prioritization of a of a homeostatic error signal is the feeling of that signal it is the rendering conscious of that signal and that is what we mean by feeling something it means now now this need is not just a need now this need is a drive now this need is driving my voluntary behavior and that is why it becomes conscious because as i said to you earlier voluntary behavior can be defined as the making of as opposed to the executing of automatized predictions just to add to uh notes on that when hearing steven's questions about spatial localization and how that can refer to brain localization as well as like in the peripersonal space it made me think about how awareness a awareness it could be without awareness awareness could be spatially dislocated or awareness might be a highly spatialized percept and then the other note that i wanted to make was um as a researcher of distributed physiology in the eu social insect colony this extended homeostatic perspective um leads to many interesting connections when we think about extended social homeostasis as well but just two notes and we can carry on very good points so you're saying we can carry on meaning i can carry on whichever way you'd like to okay great so i'm now going to move to my to my next uh my next claim uh which is which is i see we have half an hour left so i must i mustn't it's really quite amazing you know because you are so generous with giving your speakers two hours you lull us into a full sense of security and thinking well i can just elaborate as much as i want to there's plenty of time and now we're down to our last half hour and i've still got three points to go so i'm going to it's it's like uh no one goes there anymore it's too busy yeah well i'm again grateful for the time that you're giving me so let me move to this point and this has everything to do with the relationship between affective consciousness and cognitive consciousness i spoke earlier about i feel like this about that in other words the incorporating of the of the cognitive domain within the sphere of feeling and so this is the point that i'm making uh this is a a drawing that a diagram that comes from a paper that i wrote with carl fristen um and and and here um the the important equation is the third one uh and and here this diagram basically is just trying to spell out uh in in in a visual

form uh what what the what these equations are saying so when i said earlier uh that um have two ways um reducing prediction error uh uh when a when a prior prediction does not lead to the sensory state that's expected we can either change our perception in other words we can change the prediction or we can change our action we can do something differently in order to bring about the prior prediction so that's what these two equations uh describe that we have a generative model which generates predictions as to what sensory state will flow from our actions so here so here there's an action in the external world which is predicted to bring about a certain sensory state and to the extent that it does not bring about that sensory state in other words the the difference uh here is the prediction error and of course the prediction error is then used to update the generative model uh in order to give rise uh to better predictions in order to better maintain your expected sensory states i i need to emphasize here because this was not the case in the early days of the predictive processing paradigm uh that that cole fristen and and his colleagues uh uh unleashed upon the world uh century states do not necessarily mean extraceptive sensory states and so in the examples that i've been giving in this talk so far that this would be the equivalent of for example core body temperature or blood oxygen level um these uh the the most important sensory states for the organism um are its viable states in terms of its homeostatic uh expectations please remember that when i speak about actions in the world uh the world that we're talking about here uh is the can be the visceral body uh and it can be the external world the body the the mark of blanket um it it it the the the the body is as much external uh to the blanket uh as the external world is external of the nervous system um i think we've got some we're in somebody's kitchen here which you must put you you must put your mic whoever is doing their dishes there um so so uh what what this uh equation here which is which is which is uh uh written in in words here the rate of change of precision uh which is which is uh uh ω in the equation uh the rate of change of precision over time depends on how much free energy changes when you change precision this means that precision will look as if it's trying to minimize free energy the rate of this free energy minimization process is the difference between the inverse precision and the sum of squared prediction errors so that's this equation here uh put into words and the it's it's foregrounding the the central role that precision plays in minimizing prediction error at the the the i want to be clear that this precision modulation uh is in physiological terms uh the modulation of postsynaptic gain so the message passing going on here between error signals and and prediction signals is modulated posts it's it's the reticular activating systems modulation of that message passing so it's the increasing or decreasing of the gain on the error

signals that's the role of precision modulation which is just the same thing as to say that is the role of the reticular activating system it is modulating the gain uh in the message passing it to speak physiologically to speak computationally it is a matter of increasing or decreasing precision values attached to the predictions over the errors so this so to put it in in into different uh words um if things are turning out as expected that's good if uncertainty prevails that's bad so increasing confidence in a prediction is good in other words increasing precision in the prediction is good uh increasing confidence in an error signal is bad uh in other words uh the the more uncertainty the more uh you you become clear that things are not turning out as expected to that extent of course uh your confidence in your current policy uh is reduced uh in other words the precision in your current policy is reduced and that just is bad so the goodness and badness the pleasure and displeasure function of precision modulation that i described to you is it has this enormously important contribution to make to the whole of this mechanism by determining uh the influence of the error signal over the predictive model to the extent precision uh is reduced in the error signal and thereby confidence is maintained in the policy to that extent the error signal will or will not have influence over the parameters of the of the predictive model and so this is trying to illustrate the crucial role that effect plays um in in in this whole predictive mechanism so this is the this is as it were the role of the reticular activating system of the modulation of the message passing that goes on um in the these are these are as it were this synaptic transmission mechanisms and these are post-synaptic modulatory mechanisms and so i'm just wanting to link these formalisms to the affective role that the reticular activating system this so just to go back to the statement here the mechanism of perceptual and cognitive consciousness is precision modulation of allostatic prediction errors in other words it is the it is the modulation of confidence in a current policy in other words uh the the perceived consequences of a current uh so so as you as the as the a particular need is prioritized uh that need is felt this generates a category of predictive policies in other words this is what i will which is what i do in this situation in this context that i find myself in this is what i expect the consequences will be um in other words these are my expected uh and and there's an expected precision attaching to the error signals of course all the other domains of need remember what i was saying earlier here about the needs must be prioritized the other domains have monotonous precisions so the crucial mechanism in terms of voluntary action is that to use the phrase that one of you said you liked earlier um is the palpating of the uncertainty in other words the palpating of the precision and the adjustment of the confidence in the current perceptual and cognitive in other words the alistic aspect of of what i must do

about this need state because there's a state of uncertainty that's been prioritized there there's a changing of one's mind on the fly this is what this mechanism makes possible so it is the affects are as it were a drive for or a demand for predictive work for for mental work and this the modulation the palpating of the confidence in the policy over the error signals is the predictive work so demanded so in other words the work of of updating one's policy of changing one's mind it is all of it underwritten by the affect of demand so the affect of demand for predictive work what gives rise to the predictive work itself which is the palpating of uncertainties uh in the current policy and the sensory states that it gives rise to and so this is changing your mind in other words in other words voluntary action in other words the capacity for choice and this is the crucial role that precision modulation plays in that process that's uh oh okay good so i think let me just pause for a moment to see if there any questions about that because i wanted to illustrate it with the case which i think is going to pretty much take us uh to to to to the end of our time so before i launch into this last case let's see if anybody has any questions or comments about that mechanism i think we'll go to the case and then there's always so much to unpack with defining the variables and understanding what the edges mean and so on but i think the case is a good way to close this is a this is a patient of mine mr s and he had a meningioma here at the base of his frontal lobes it was an olfactory sheath meningioma um and it pushed on his optic nerves and as a result of that um he his vision was was impaired and this is how the tumor came to attention and it was successfully surgically resected now because of its location in relation in fact to the optic nerves there was some nervousness on the part of the surgeon to remove it in its entirety in fact he felt it was not possible to remove it in its entirety so he left a little nub of the tumor and uh this regrew and so uh the the the patient again uh noticed these visual difficulties uh returned to the surgeon and the operation had to be repeated because of scar tissue the second operation is always trickier than the first and unfortunately in the second operation um there was a bleed and that bleed was into the basal forebrain nuclei small bleed but the basal forebrain nuclei are crucial these are the upper end of the of the remember i was saying earlier that these different parts of the ascending arousal mechanisms of the of the brain that they that they are the source nuclei for these different neuromodulators and i spoke of dopamine and noradrenaline and serotonin earlier these nuclei are the source nuclei or or there are other source nuclei for acetylcholine here too but these are very important source nuclei for acetylcholine now remember all of these what i was saying earlier about these neuromodulatory systems is that they are modulating postsynaptic gain which is just the same thing physiologically as to say

computationally they're modulating precision that's what they do they up and down regulate the precision uh in the message passing and this as i said earlier it dictates you know which which messages are going to be selected and and which not so it's a crucially important role uh does precision modulation um in cognition and um and i'm now wanting to show you this relationship the relationship between the affective mechanisms that i'm talking about that is the the affective functions that are performed centrally by periaqueductal gray and then how this gives rise to the modulation through these arousal systems of these different neurotransmitter systems and i'm going to now show you how this worked in this case acetylcholine modulates the confidence in error signals let me just quickly tell you a little bit more about this patient um he was he was 56 years old at the time of the second operation and as a result of the damage to the basal forebrain nuclei from the second operation he woke up from the surgery with a condition called confabulatory amnesia so although the tumor was resected successfully and the visual problem was was corrected he now had this devastating new condition uh called confabulatory amnesia confabulatory amnesia the patient is it's not only that they are um amnesic in other words that they're unable to remember particularly recent events uh but there's also quite a long retrograde extension too in other words there's quite an impairment of their retrograde memory too not only do they have this but they also are not aware of their memory problems and so when they draw up um when they attempt to retrieve a memory their the memory that they retrieve is not the correct one and they don't realize it's not the correct one and so uh they have what appear to be false memories these are the confabulations they frequently are related um in in in some semantic sense with the target memory with the memory that they're looking for um there's some semantic relationship between the memory they find and the one they're looking for but they can be grossly mis misplaced in space and time and so these are what we call confabulations and the patient is not sufficiently critical of these misrememberings as a result of the damage to these neuromodulatory mechanisms so uh this patient mr s just to give you one example i mean an extreme example um he uh he had his operation in johannesburg uh which is in south africa which is where uh i hail from but at that at the time i was living and working in london and so the surgeon who i knew well sent the patient to see me to consult me in london um because i was doing work with this with this condition confabulatory amnesia so the patient arrives in london on a friday and comes to me on the monday and has no idea that he's in london because of course he doesn't remember the journey he can't remember anything from one minute to the next and so uh i say something about

well you know the the the the surgeon whose name was mr miller the surgeon referred you to me because of this memory difficulty that you're having that's why he sent you to london and he said london uh what do you mean london and i said yes you know you're in london you don't realize it because you don't remember the journey uh you know that's the whole point this is the kind of problem that you're having and um he denies it he says he's not in london now it's that so that's the confabulatory aspect that these patients in his memory uh he's in johannesburg and so he believes he is in johannesburg so i point out to him it was winter and it was it was snowing outside i point out to him uh the the the wintry conditions outside you never by the way have snow in johannesburg so i said look out the window he looks art is absolutely shocked but then retorts no i know i'm in johannesburg just because you're eating pizza doesn't mean you're in italy that's what he says to me so in other words you know you don't have to you know you mustn't overrate the evidence of your senses and so that's just an extreme example of what i mean by how how these patients uh how how uh how their their amnesia it's not just a lack of memory it's also an excessive confidence uh in the in the incorrect memories that they draw up so uh that's the that's the background and and i saw this man the whole point of him being referred to me is because i treat such patients and so i i then saw him six days a week um if at the same time in my outpatient clinic at the royal london hospital same time same place he came with his wife to the waiting room i would then go collect him take him up to my consulting room uh spend an hour with him take him back down to the waiting room and then i would talk to his wife uh because uh he he he was so full of uh confabulations i needed to verify things um in order to get some of what was going on and there were certain themes and this is an interesting thing about these patients the these confabulatory patients there's certain themes uh that returned again and again uh the patient was an electronic engineer but but he uh in reality he was but he thought the reason he was coming to me frequently he thought he was coming to me to to to because i was consulting him about an electronic problem or otherwise he thought that the two of us were electronic engineers together working on some electronic problem also frequently he thought that he and i were in in some sporting team together that we played rugby together uh or or we we were in this a rowing team together now his wife told me he had played rugby uh at university uh a good 30 years or more before and likewise he had he had been a keen rower but this too was at university more than 30 years before and so these are again uh good examples of confabulations you know the patient mislocating in space and time a memory that that he draws up now and he has too much confidence in that he's too too readily um accepts uh

veracity of the products of his own memory search so that's the background now on this particular day that i want to give you a little snippet of a session from uh he on this day when i came to the waiting room he touched the scar on his head the craniotomy scar on his head and he said hi doc to me so this was progress you know for the first time he was associating me with medicine and he was associating me with the surgical scar on his head and so i thought this was great progress and so when we got into my consulting room i said to him you touched your head when we met in the waiting room and he said i think the problem is that a cartridge is missing we must we just need the specs what was it a c-49 should we order it so i said to him what does a c-49 cartridge do he says memory it's a memory cartridge a memory implant oh sorry i should have mentioned this his wife told me that he had had dental implants he'd had serious problems with his teeth and these had been for many years but the these problems had finally been successfully treated uh by implants uh teeth implanted into the into the jaws so when he said that that c49 cartridge is a memory cartridge and that it's a memory implant it brought that operation that dental operation to my mind he said but i never really understood it in fact i haven't used it for a good five or six months now it seems we don't really need it it was all chopped away by a doctor what's his name dr solms i think but it seems i don't really need it the implants work fine so i said to him you're aware that something's wrong with your memory but and he interrupts me and he says yeah it's not working a hundred percent but we don't really need it it was just missing a few beats the analysis showed there was some c or c09 missing denise brought me here to see a doctor what's his name again dr psalms or something and he did one of those heart transplant things so this is referring to another operation that he'd had which is clearly also being referred to here which was that he had a cardiac arrhythmia and so he had a pacemaker fitted a cardiac pacemaker so he says so he did one of those heart transplant things and now it's working fine never misses a beat so i said to him which is what i actually thought i said you're aware that something's amiss some memories are missing and of course that's worrying you hope i can fix it just like those other doctors fix the problems with your teeth and your heart but you want it so much that you're having difficulty accepting that it's not fixed already so he goes on and says oh i see yes it's not working 100 and he touches his head he says i got knocked on the head went off the field for a few minutes but it's fine now i suppose i shouldn't go back on but you know me i don't like going down so i asked tim knox tim knox by the way is a sports physician in south africa he says so i asked him nurks because i've got the insurance you know so why not use it why not go to the best and he said fine play on the that's the end of the

snippet that i wanted to show you um by the way oh before i to show you about that so so this is a case uh in which due to damage to one of the precision modulating nuclei or sets of nuclei in this man's case the basal forebrain due to the damage to the acetylcholine modulating mechanisms that he has too much confidence in his predictions uh and too little confidence in the he does not up the gain uh on the arizona say looks i say to him his predictive model tells him he's in johannesburg i said to him no you're in london look out the window he looks out the window he sees snow you know clearly not something that could possibly be associated with johannesburg but he sticks with his prediction he says no i know i'm in joburg uh just because you're eating pizza it doesn't mean you're in italy uh and likewise uh in the case of the these memory processes that i showed you in that session each time uh that he starts to feel uh the unpleasure uh of the mounting error signals the mounting the the things are so just think about it he has he touches his head he's on the brink of being aware that he's he's had a brain operation uh that the brain operation has resulted in loss of memory i i hope you can see that with him saying uh that uh i i he says we just need to order the specs for this module that's missing what does the module do it does memory you know so touching his head uh speaking about operations so he's on the he's on the brink of recognizing that things were not as he expected that things are in fact uh quite different my point being that this evokes feelings this is bad this is a this is a panic inducing situation and because this man has damage to these precision modulation mechanisms that we've been talking about um that he what he does is he simply up regulates the or maintains his confidence in his prediction and in this way maintains his his emotional equanimity rather than allowing the unpleasant uh uh affect to dominate and the unpleasant unpleasant effect uh to update um to to to to upregulate the error signal uh and which would which would normally update the predictive model so i'm hoping that in this case you can see something of the role of affect uh that the role that the feelings involved in this man's case uh in relation to uh his predictions uh and the prediction errors in other words the cognitive business um of what he perceives uh and what he believes and how what he perceives changes what he believes and the role of feeling in all of this i thought this was a succinct case to be able to illustrate um illustrate all of that the because it's only one case i thought i should just show you some of this several papers that we've written uh on showing how uh these mechanisms work in confabulation uh we it's not just this one case we were able that affect regulation uh that that the confed that confabulation has a wishful quality in other words it has a down regulating of error signals quality and an up-regulating of predictions quality and we were we were also able to show uh that by analysis of transcripts of cases like

we studied many cases like this how the affect actually improves uh with each confabulation so there's a increasing negative effect followed by confabulation uh followed by an improvement in the affect so um i see i'm right at the end of my time i just want to uh i don't have time to go into this point i'm sorry um this is this is not such a a fundamental point it's just a slightly different way that i see the predictive hierarchy um if we if we um take seriously um that what we're talking about here um fundamentally uh we're talking about homeostatic systems and that the most important predictions have to do with maintaining your the your phenotypic prior preference distribution that this has implications for how we conceptualize the predictive hierarchy but as i say we don't have time to go into that and then the last thing is just to say that these models that we that i've i've together with my colleagues derived from the study of the neuroscientific evidence um on the basis of this well-known uh statement of of uh richard feynman's what i can't create i do not understand uh we we are trying to instantiate these mechanisms that are described to you earlier in an artificial consciousness um i'm working with a a group of really great guys physicists and and computer scientists and applied mathematicians and we presented our uh our preliminary findings a a few months ago at um uh carl fristan's theoretical neurobiology meetings um and so watch the space um we are we're we are saying that uh if we have identified the mechanism uh the causal mechanism where whereby affects are generated what they actually do in relation to a self-organizing system then if this really is how affect is is generated if this really is how it is caused then we should be able to instantiate it uh in a in an artificial system and and that's what we're what we're trying to do at the moment i'm sorry to have rushed at the end um if you'll indulge me i will wait for if there are any questions comments um i just wanted to show you my so this is i'm saying these questions can be that they look quite different uh if we look at it through the lens that i've tried to encourage you to look at it with me through in other words through the lens of affect these questions why and how is this something it's like to be an organism something it's like for the organism and questions like why is the performance of these functions accompanied by experience this is thomas's question why doesn't all this information processing go on in the dark free of any in the field i'm saying that these questions i've tried to provide you a clue as to how we might be able to address them differently um oh what happened there i've lost my oh oh my good ah there i am yes how we might be able to address them differently but get them through this lens um sorry to have gone on my word i can't believe i've spoken for two hours again but if you'll if you will allow me i'll be happy to wait for any further comments and questions before we end as as we have actually a whole

document of questions and many questions in the chat i think it's a perfect place to um end the live session and whenever the affordance presents itself we're always happy to have a dot three the numbers keep counting and we'll keep hosting the sessions because this was very fascinating and i hope people got a lot out of it and very tantalizing thank you so much for inviting me and apologies for going on for so long and i really enjoyed what little interchange we had over these four hours i really enjoyed thank you so much maybe one nice um future uh event could be picking up on those last two which are extremely interesting and important and then starting with those two with the dot one and the dot two as prerequisite and then sharing more about those two um when the time is right and having that as more of a full discussion very good thank you mark thank you stephen dave thanks steven thanks steve thanks daniel peace out take care bye all right thanks everybody for watching hope you enjoyed that see you later